

WEBSITE TRAFFIC ANALYSIS

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Objective

The objective of this project is to analyze website traffic data to understand user engagement patterns, event activity, geographical distribution, and opportunities to improve website conversions.

Dataset Information

- Dataset Name: Website Traffic Analysis Dataset
- Source: Kaggle

Tools Used:

- Python
- Pandas
- Matplotlib

The dataset contains user interaction events such as page views, clicks, previews, and location-based information.

Data Cleaning and Preparation

The following data preparation steps were performed:

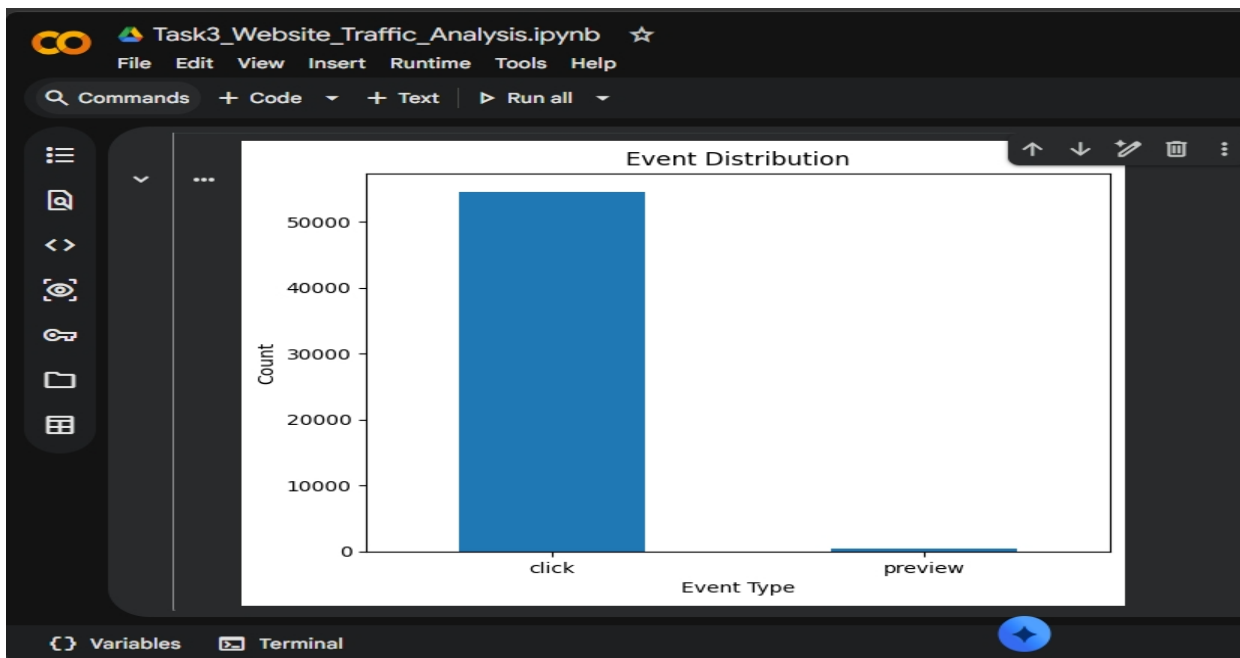
- Checked dataset structure using `df.info()`
- Identified missing values using `df.isnull().sum()`
- Reviewed event distributions
- Analyzed country and city level traffic
- Prepared data for visualization

Website Traffic Metrics

Event Summary	
Page Views	142,015
Clicks	55,732
Previews	28,531

Visualizations:

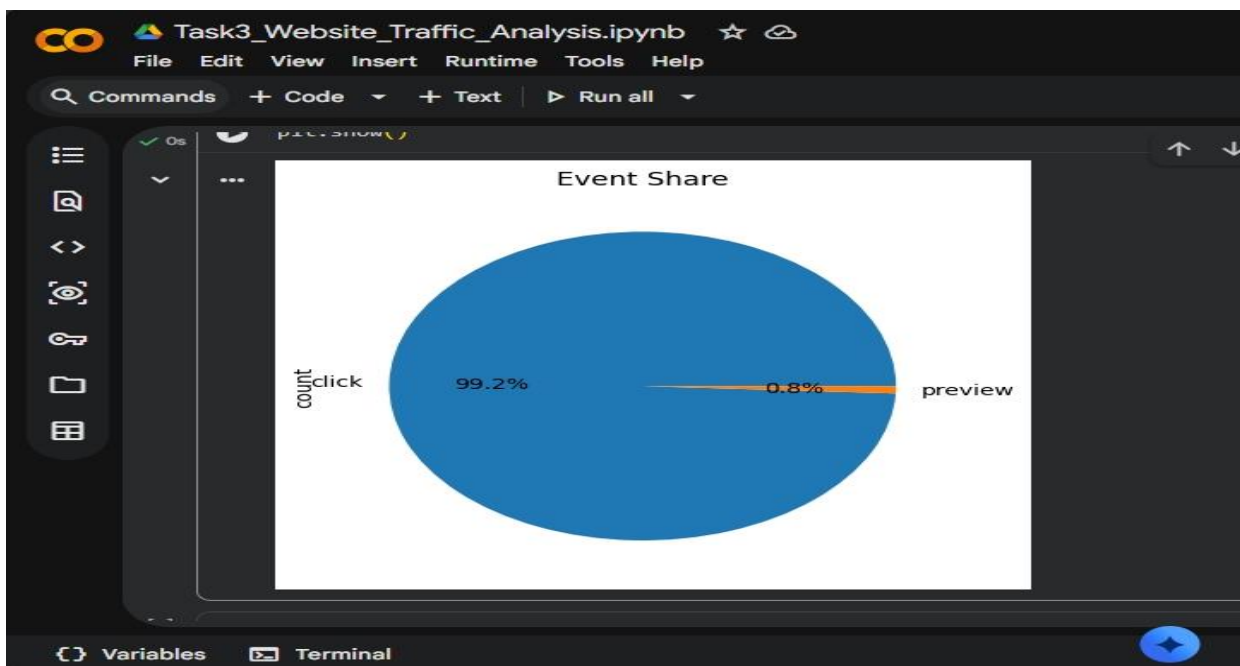
Event Distribution



Observation:

Page views represent the highest user activity on the website.

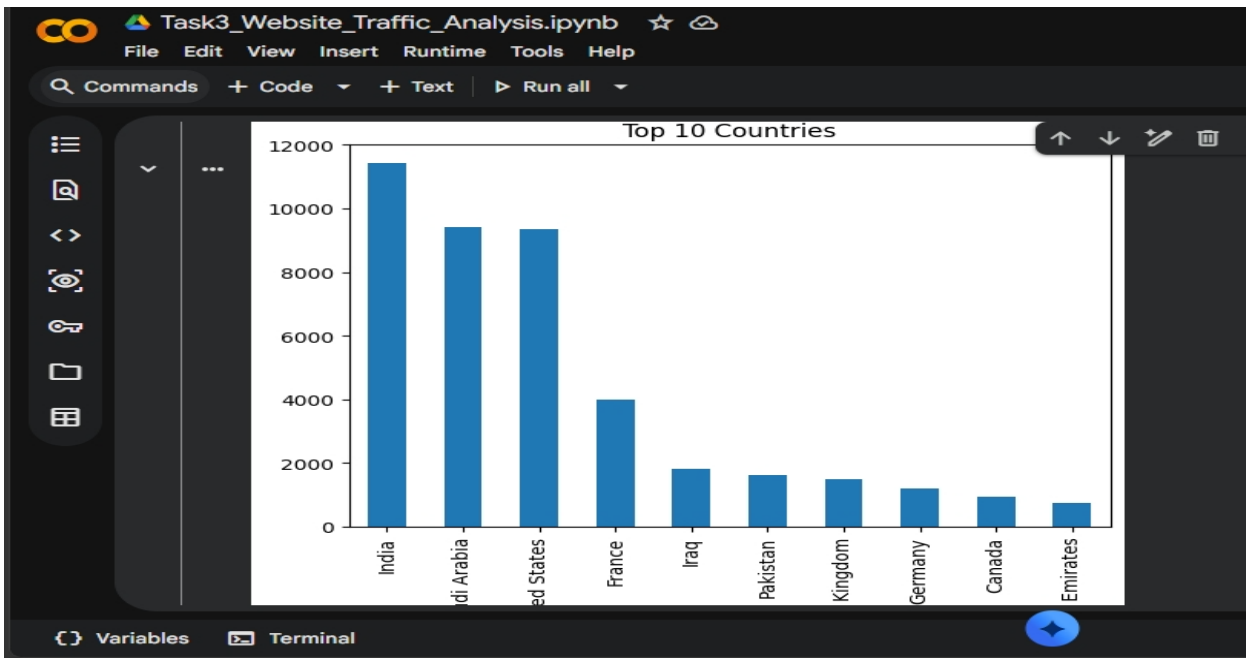
Event Share



Observation:

Page views account for the largest percentage of website interactions.

Top 10 Countries

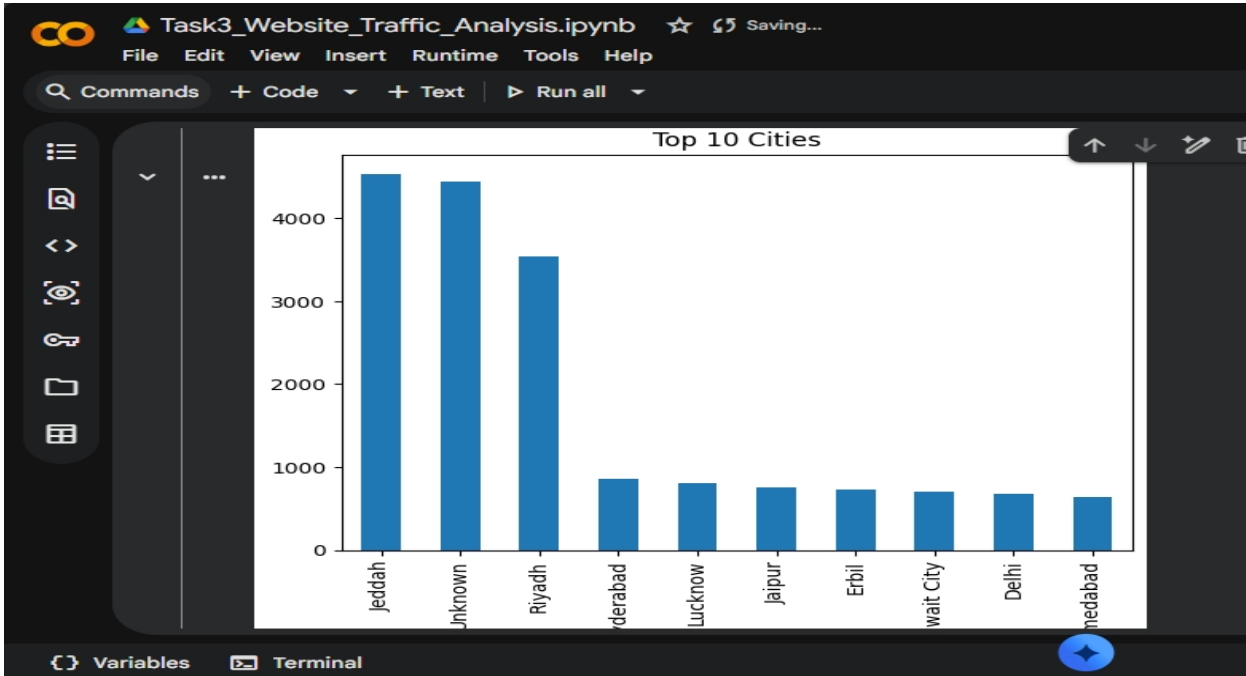


Observation

Observation :

India generated the highest website traffic, followed by Saudi Arabia and the United States.

Top 10 Cities



Observation:

Jeddah and Riyadh contributed significant traffic among all cities.

Geographic Analysis

Top Countries	Traffic
India	11,429
Saudi Arabia	9,409
United States	9,366
France	3,993
Iraq	1,823

Top Cities	Traffic
Jeddah	4,528
Unknown	4,444
Riyadh	3,535
Hyderabad	854
Lucknow	804

Key Findings

- 1. Page views account for the majority of website activity.
- 2. Click events indicate strong user engagement.
- 3. Preview events represent customer interest before action.
- 4. India is the largest traffic source.
- 5. Traffic is concentrated in a few countries and cities.

Recommendations

- 1. Improve call-to-action buttons to increase click-through rates.
- 2. Optimize high-traffic landing pages.
- 3. Personalize content for visitors from top-performing countries.
- 4. Improve user experience on pages with high preview activity.
- 5. Implement targeted marketing campaigns to increase conversions.

Conclusion

The website traffic analysis revealed strong user engagement through page views, clicks, and previews. India emerged as the leading traffic source, while several cities contributed significant visitor activity. By optimizing landing pages and improving conversion strategies, Alfido Tech can enhance website performance and user engagement.